

Network Structure of Wikipedia Attention and Financial Market Volatility

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Abstract

This study investigates how the network structure of attention on Wikipedia influences financial market volatility, focusing on NVIDIA (NVDA) stock during 2022–2025. Using daily pageview data for ten AI-related Wikipedia pages ($N=790$ trading days), we test whether network topology—measured by weighted eigenvector centrality—predicts market volatility beyond simple attention volume. Employing time-series regression with Newey-West standard errors, we find that network centrality significantly predicts increased volatility ($\beta = 0.062$, $p < 0.01$), while total pageview volume shows no effect. Shannon entropy exhibits positive associations with volatility, whereas the Herfindahl-Hirschman Index shows negative relationships. Attention metrics display strong endogenous persistence ($R^2=0.987$), while volatility persistence is limited in our extended sample. The DeepSeek announcement (January 21, 2025) shows no significant event effect. These findings demonstrate that *how* information diffuses through network structures matters more than *how much* attention topics receive, contributing to diffusion theory and behavioral finance literature.

Keywords: information diffusion, network centrality, Wikipedia, financial markets, volatility, time series analysis

1 Introduction

The digital transformation of information ecosystems has fundamentally altered how knowledge spreads through society. Wikipedia, as one of the world’s largest collaborative knowledge repositories, provides a unique window into collective attention dynamics and information diffusion patterns (23). With over 60 million articles and billions of monthly

pageviews, Wikipedia serves as both a knowledge platform and a real-time indicator of public interest in specific topics. Understanding how attention flows through Wikipedia’s network structure has become increasingly relevant as digital platforms mediate an ever-larger share of information consumption.

Financial markets represent a domain where information diffusion may have measurable consequences and research demonstrates that online attention, measured through search queries or social media activity, correlates with market behavior (6; 3). However, most studies focus on attention *volume* rather than the *structure* of attention diffusion through information networks.

The case of NVIDIA Corporation (NVDA) is particularly pertinent for investigating these dynamics. As the leading manufacturer of graphics processing units (GPUs) essential for artificial intelligence applications, NVIDIA has been at the epicenter of the AI revolution since 2022. The company’s stock price has exhibited extraordinary volatility, reflecting both technological breakthroughs and evolving public discourse about AI’s implications. Wikipedia pageviews for AI-related topics thus provide a proxy for public attention to this transformative technology and its commercial manifestations.

1.1 Social and Scientific Relevance

1.1.1 Social Relevance

The social relevance of this research emerges from the growing influence of digital information ecosystems on economic behavior. As financial markets increasingly respond to online information flows rather than traditional media, understanding these relationships helps multiple stakeholders navigate information-rich environments. Individual investors can benefit from recognizing how attention patterns signal market uncertainty and on the other hand policymakers concerned with market stability need to understand how digital platforms mediate information that moves markets. Platform designers bear responsibility for how their architectural choices structure information flows with real economic consequences.

Moreover, the AI sector’s rapid growth—with market capitalizations in the trillions—means that information diffusion about AI technologies directly affects substantial economic value (5). Understanding how attention to AI concepts flows through knowledge networks and influences investor behavior thus addresses a socially significant phenomenon affecting millions of investors and broader economic stability.

1.1.2 Scientific Relevance

The scientific relevance resides in bridging communication science theories on information diffusion with financial econometrics. While information diffusion theory and net-

work theory offer valuable frameworks for understanding social structures and positions, their integration to specifically examine market outcomes can lead to discover particular patterns.

Existing research has established relationships between online attention and financial indicators (17; 20), but some knowledge gaps remain:

1. **Network vs. Volume:** Prior studies typically measure attention as simple pageview counts or search volumes. Whether the *network structure* of attention—which pages are central vs. peripheral—matters independently of volume remains unexplored.
2. **Endogenous vs. Exogenous Dynamics:** Research has not systematically distinguished between self-reinforcing attention patterns (endogenous processes) and external shock-driven changes (exogenous events) in the context of Wikipedia-market relationships.
3. **Mechanism Specification:** The theoretical mechanism linking Wikipedia attention structure to market volatility requires clarification. Does centralized attention amplify market reactions, or does dispersed attention signal broader uncertainty?

1.2 Research Questions

This study addresses the following research questions:

1. **RQ1:** To what extent does the network centrality of Wikipedia pages predict NVIDIA stock volatility, independent of total attention volume?
2. **RQ2:** How do alternative measures of attention structure—entropy (dispersion) and HHI (concentration)—relate to market volatility?
3. **RQ3:** Do attention metrics exhibit endogenous persistence over time, and does volatility display similar autocorrelation?
4. **RQ4:** Does the DeepSeek announcement (January 21, 2025) represent a significant exogenous shock to the attention-volatility relationship?

2 Theory and Hypotheses

2.1 Information Diffusion Theory

Information diffusion theory provides the foundational framework for understanding how ideas, knowledge, and innovations spread through social systems. The theory identifies four key elements: (1) the innovation or information itself, (2) communication channels

through which information flows, (3) time over which diffusion occurs, and (4) the social system within which diffusion takes place. In digital environments, these elements transform: communication channels become platform architectures, time compresses due to instant connectivity, and social systems manifest as online communities and networks (9).

Wikipedia’s collaborative nature makes it an ideal platform for studying information diffusion. Pageviews represent quantifiable attention that diffuses through the platform’s network structure of interconnected articles (22). Unlike social media platforms where diffusion follows social network ties, Wikipedia diffusion follows the semantic network of hyperlinked knowledge topics. This structural difference is theoretically significant: attention flows through conceptual relationships rather than personal connections.

The two-step flow model (11) suggests that information often reaches the mass public through opinion leaders or influential nodes rather than directly from sources. Translating this to Wikipedia’s context, pages with high structural centrality may serve as such influential nodes, channeling attention toward specific topics and potentially amplifying their salience. When major events occur—such as technological breakthroughs in AI—attention may concentrate on these central nodes, creating cascading effects through the information network.

2.2 Network Theory and Structural Position

Network theory provides analytical tools for understanding structural positions within interconnected systems. Centrality measures quantify a node’s importance within a network, with different metrics capturing distinct aspects of structural position. Eigenvector centrality, specifically, identifies nodes that are connected to other well-connected nodes, positioning them as influential within the overall network structure (4).

When applied to Wikipedia’s knowledge network, eigenvector centrality identifies pages that occupy strategic positions in information flows. A page with high eigenvector centrality not only receives direct attention but is also connected to other pages that themselves receive significant attention. This creates a multiplier effect: information reaching a central page has higher potential to diffuse widely through the network.

We hypothesize that when attention concentrates on pages with high eigenvector centrality, this structurally privileged position amplifies attention diffusion through the broader information ecosystem. For financial markets processing AI-related information, such concentrated attention on central nodes may signal heightened salience of particular concepts, increasing market participants’ uncertainty and so, volatility.

This network amplification perspective differs from volume-based attention theories. Simple pageview counts treat all attention equally, regardless of where it occurs in the network structure. Network-based measures recognize that attention to structurally cen-

tral topics may have disproportionate influence on information processing and decision-making.

2.3 Endogenous Persistence Processes

Time-series dynamics in both information diffusion and financial markets exhibit endogenous persistence—patterns that perpetuate themselves through feedback loops (10). In financial markets, volatility clustering represents a well-documented phenomenon: periods of high volatility tend to be followed by continued high volatility (13; 8). This persistence arises from behavioral factors (risk aversion patterns) and mechanical effects (stop-loss cascades, margin calls) that create self-reinforcing dynamics.

Information attention may exhibit similar persistence through cumulative advantage processes (15). Pages that receive attention become more visible through platform algorithms and user behavior patterns, attracting additional attention in a positive feedback loop. Network positions reinforce themselves: central pages remain central because their structural position continues to channel attention toward them.

These endogenous processes generate the following hypotheses:

H1a (Endogenous): NVIDIA stock volatility exhibits positive autocorrelation, such that high volatility on day $t - 1$ predicts high volatility on day t , reflecting endogenous market dynamics.

H1b (Endogenous): Wikipedia attention metrics (centrality, entropy, HHI) exhibit positive temporal dependence, such that yesterday’s attention structure predicts today’s attention structure, reflecting endogenous information diffusion dynamics.

2.4 Exogenous Amplification and Dispersion

While endogenous processes explain baseline persistence, exogenous relationships between attention structure and market outcomes operate through distinct mechanisms. Network amplification theory suggests that attention concentrated on structurally central nodes has greater influence on external outcomes than attention distributed across peripheral nodes (21).

When information about AI developments reaches Wikipedia, users navigating this information ecosystem encounter pages with varying structural positions. Attention to highly central pages—those connected to many other relevant topics—may signal broader salience and implications. For market participants monitoring AI developments, observing attention concentrated on central Wikipedia pages (e.g., “Artificial intelligence,” “Deep learning”) may trigger greater uncertainty about technological trajectories and competitive dynamics, manifesting as increased trading volatility.

Conversely, attention dispersion can be understood through information entropy theory (19). Shannon entropy measures disorder in probability distributions; applied to Wikipedia pageviews, high entropy indicates attention spread across many pages rather than concentrated on few. High entropy may signal diffuse uncertainty—market participants uncertain about which AI concepts matter most—potentially increasing volatility through disagreement among traders.

The Herfindahl-Hirschman Index (HHI), traditionally used to measure market concentration (18), can similarly measure attention concentration. Higher HHI indicates attention focused on fewer pages. We hypothesize this concentrated focus may actually *reduce* volatility by creating clearer consensus about which developments matter, whereas dispersed attention (low HHI) reflects uncertainty.

These considerations yield the following hypotheses:

H2a (Exogenous): Higher weighted network centrality of Wikipedia pages is associated with increased NVIDIA stock volatility, independent of total attention volume, reflecting network amplification effects.

H2b (Exogenous): Higher entropy (attention dispersion across pages) predicts increased market volatility, reflecting diffuse uncertainty.

H2c (Exogenous): Higher HHI (attention concentration on fewer pages) predicts decreased market volatility, reflecting focused attention and reduced uncertainty.

2.5 Event-Driven Information Shocks

Exogenous events can disrupt normal information diffusion patterns, creating abnormal information flows that affect both attention structures and market behavior (1). Event study methodology in finance (12) examines how specific announcements generate abnormal returns or volatility around event dates. In communication science, agenda-setting research (14) demonstrates how prominent news events shift public attention patterns.

The January 21, 2025 DeepSeek announcement—where a Chinese AI laboratory released a competitive large language model—represents a potential exogenous shock within the AI ecosystem. Such announcements may trigger attention spikes on Wikipedia as users seek information about new developments, simultaneously affecting investor perceptions of competitive dynamics in the AI sector.

However, detecting event effects in extended time series presents methodological challenges. A single day’s effect may be statistically undetectable in a sample of 790 days, requiring careful specification of event windows and consideration of whether effects concentrate on announcement dates or diffuse across longer periods.

H3 (Exogenous): The DeepSeek announcement (January 21, 2025) generates a significant increase in both Wikipedia attention centrality and NVIDIA volatility, representing an exogenous information shock.

3 Data and Methods

3.1 Research Design

This study employs a longitudinal time-series design analyzing daily observations from January 1, 2022 through February 28, 2025. This three-year window captures the rapid evolution of public interest in AI following ChatGPT’s November 2022 release while providing sufficient observations (N=790 trading days) for robust time-series analysis. The design enables examination of both endogenous persistence patterns and exogenous relationships between Wikipedia attention structure and NVIDIA stock volatility.

3.2 Data Collection

3.2.1 Wikipedia Pageview Data

We collected daily pageview data for ten Wikipedia pages using the Wikimedia REST API (https://wikimedia.org/api/rest_v1/). Page selection followed three criteria:

1. **Relevance:** Pages must relate to artificial intelligence and NVIDIA’s business ecosystem
2. **Coverage:** Pages must have consistent data availability throughout the study period
3. **Network structure:** Pages must exhibit semantic relationships enabling network construction

Table 1 presents the selected pages with rationales for inclusion.

Data collection proceeded through the following steps:

1. **API requests:** For each page, daily aggregated pageviews were requested using the `httr` package in R, specifying date ranges in YYYYMMDD format required by the API.
2. **Data parsing:** JSON responses were parsed using `jsonlite`, extracting timestamp and pageview count fields. Dates were converted to R Date objects for proper temporal sorting.

Table 1: Wikipedia Pages Analyzed

Page Title	Rationale for Inclusion
NVIDIA	Primary company of interest; leading GPU manufacturer for AI applications
Artificial_intelligence	Broad conceptual category encompassing all AI topics; central node in knowledge network
OpenAI	Leading AI research organization; developer of ChatGPT and GPT models
ChatGPT	Breakthrough AI application that catalyzed public attention to AI (Nov 2022)
Large_language_model	Core technology behind recent AI advancements; technical foundation
Machine_learning	Foundational AI methodology; established field pre-dating recent breakthroughs
Deep_learning	Key technique central to modern AI and NVIDIA’s GPU applications
Semiconductor	Industry category including NVIDIA’s core hardware products
GPU	Specific hardware specialization central to NVIDIA’s business and AI training
Transformer_(ML_model)	Architectural innovation enabling large language models

- Missing data handling:** Three pages (Large_language_model, ChatGPT, Transformer) showed missing data for periods before their Wikipedia article creation dates. For these pages, zero values were assigned to pre-creation periods, as these topics genuinely had no Wikipedia presence during those times.
- Wide format transformation:** Data were transformed from long format (page-day observations) to wide format (days with columns for each page) using `tidyr::pivot_wider()`, facilitating multivariate analysis.

The final Wikipedia dataset contains 1,155 calendar days (Jan 1, 2022 – Feb 28, 2025) $\times$ 10 pages = 11,550 page-day observations. This substantially exceeds the course requirement of 1,000 observations for multiple time series analysis.

3.2.2 Network Construction

A directed network was constructed based on semantic relationships between Wikipedia pages. Edge creation followed these principles:

- Technological dependency:** Edges connect pages where one concept depends on another (e.g., ChatGPT $\rightarrow$ Large_language_model, as ChatGPT is built on LLM technology)

2. **Industry relationships:** Edges connect pages where business/product relationships exist (e.g., NVIDIA → GPU, as NVIDIA manufactures GPUs)
3. **Conceptual hierarchy:** Edges connect specific topics to broader categories (e.g., Deep_learning → Machine_learning, as deep learning is a subset of ML)

The network was constructed using the **igraph** package in R. Network statistics: 10 nodes (pages), 27 directed edges, network density = 0.30, average degree = 5.4. Figure ?? would visualize this structure (not included due to space constraints; available in supplementary materials).

3.2.3 Financial Data

Daily adjusted closing prices for NVIDIA (NVDA) were obtained from Yahoo Finance using the **quantmod** package in R. The **getSymbols()** function retrieved historical price data for the matching period (Jan 1, 2022 – Feb 28, 2025). Financial markets operate only on weekdays excluding holidays; thus, the financial dataset contains 791 trading days compared to 1,155 calendar days in the Wikipedia data.

Daily volatility was computed as absolute logarithmic returns:

$$\text{Volatility}_t = \left| \ln \left(\frac{P_t}{P_{t-1}} \right) \right| \quad (1)$$

where P_t represents the adjusted closing price on trading day t . This measure is standard in financial econometrics and captures price fluctuation magnitude while being scale-invariant (7).

3.3 Variable Construction

3.3.1 Dependent Variable

Volatility (volatility_t): Absolute logarithmic daily returns for NVDA stock, as defined above. This continuous variable measures market uncertainty and information processing on each trading day.

3.3.2 Independent Variables

Weighted Network Centrality (cent_t): Daily eigenvector centrality weighted by pageview shares. For each day t , we computed:

$$\text{cent}_t = \sum_{i=1}^{10} \left(\frac{\text{views}_{it}}{\sum_{j=1}^{10} \text{views}_{jt}} \right) \times \text{eigenvector}_i \quad (2)$$

where views_{it} represents pageviews for page i on day t , and eigenvector_i is the eigenvector centrality score for page i in the static network (computed once using `igraph::eigen_centrality()`). This measure captures where attention flows within the network structure, giving greater weight to days when attention concentrates on structurally central pages.

Shannon Entropy (entropy_t): Measures dispersion of attention across pages. For each day t :

$$\text{entropy}_t = - \sum_{i=1}^{10} p_{it} \ln(p_{it}), \quad \text{where } p_{it} = \frac{\text{views}_{it}}{\sum_{j=1}^{10} \text{views}_{jt}} \quad (3)$$

Higher entropy indicates attention dispersed across many pages; lower entropy indicates concentrated attention. To avoid $\ln(0)$, we added a small constant (10^{-12}) to probabilities.

Herfindahl-Hirschman Index (hhi_t): Measures concentration of attention. For each day t :

$$\text{hhi}_t = \sum_{i=1}^{10} p_{it}^2 \quad (4)$$

HHI ranges from 0.1 (perfectly dispersed: each page receives 10% of views) to 1.0 (perfectly concentrated: one page receives 100% of views).

Total Volume (volume_t): Aggregate daily pageviews across all ten pages, measuring overall attention intensity.

Event Variables: Binary indicators for testing H3:

- event_deepseek_t : Equals 1 on January 21, 2025; 0 otherwise
- event_window_t : Equals 1 during January 16–26, 2025 (± 5 trading days); 0 otherwise

3.3.3 Control Variables

Lagged Variables: For all attention metrics and volatility, we constructed one-day lags ($t - 1$) to test endogenous persistence (H1a, H1b) and control for autocorrelation:

- vol_lag1_t : Volatility on day $t - 1$
- cent_lag1_t : Weighted centrality on day $t - 1$
- entropy_lag1_t : Entropy on day $t - 1$
- hhi_lag1_t : HHI on day $t - 1$
- volume_lag1_t : Total volume on day $t - 1$

After merging Wikipedia and financial data by date (inner join to retain only trading days) and constructing lag variables, the final analytical dataset contains $N=790$ observations (trading days from January 4, 2022 through February 27, 2025).

3.4 Analytical Approach

We employ Ordinary Least Squares (OLS) time-series regression with Newey-West heteroskedasticity and autocorrelation consistent (HAC) standard errors (16). The Newey-West correction addresses two common issues in time-series data: (1) heteroskedasticity (non-constant error variance), and (2) autocorrelation (correlation between errors across time). We specify a lag parameter of 5, allowing for potential autocorrelation up to one week (5 trading days).

3.4.1 Model Specifications

To test our hypotheses, we estimate six models:

Model 1 (H1a, H2a): Tests volatility persistence and network amplification effect

$$\text{volatility}_t = \alpha + \beta_1 \text{vol_lag1}_t + \beta_2 \text{cent}_t + \beta_3 \text{cent_lag1}_t + \epsilon_t \quad (5)$$

Model 2 (H2b): Tests entropy-volatility relationship

$$\text{volatility}_t = \alpha + \beta_1 \text{vol_lag1}_t + \beta_2 \text{entropy}_t + \beta_3 \text{entropy_lag1}_t + \epsilon_t \quad (6)$$

Model 3 (H2c): Tests HHI-volatility relationship

$$\text{volatility}_t = \alpha + \beta_1 \text{vol_lag1}_t + \beta_2 \text{hhi}_t + \beta_3 \text{hhi_lag1}_t + \epsilon_t \quad (7)$$

Model 4 (H1b): Tests attention persistence (centrality as dependent variable)

$$\text{cent}_t = \alpha + \beta_1 \text{cent_lag1}_t + \beta_2 \text{entropy_lag1}_t + \beta_3 \text{vol_lag1}_t + \epsilon_t \quad (8)$$

Model 5 (H3): Tests event effects

$$\begin{aligned} \text{volatility}_t = \alpha + \beta_1 \text{vol_lag1}_t + \beta_2 \text{cent}_t + \beta_3 \text{cent_lag1}_t + \\ \beta_4 \text{event_deepseek}_t + \beta_5 \text{event_window}_t + \epsilon_t \end{aligned} \quad (9)$$

Model 6 (Full/Horse Race): Tests all attention measures simultaneously

$$\begin{aligned} \text{volatility}_t = \alpha + \beta_1 \text{vol_lag1}_t + \beta_2 \text{cent}_t + \beta_3 \text{cent_lag1}_t + \\ \beta_4 \text{entropy}_t + \beta_5 \text{entropy_lag1}_t + \beta_6 \text{hhi}_t + \beta_7 \text{hhi_lag1}_t + \\ \beta_8 \text{volume}_t + \beta_9 \text{volume_lag1}_t + \epsilon_t \end{aligned} \quad (10)$$

3.4.2 Model Diagnostics

Prior to estimation, we verified key assumptions:

- **Stationarity:** Augmented Dickey-Fuller tests confirmed all variables are stationary (all $p < 0.01$), avoiding spurious regression problems.
- **Multicollinearity:** Variance Inflation Factors (VIF) for all models remain below 3.0, well under the conventional threshold of 10, indicating no problematic multicollinearity.
- **Autocorrelation:** Ljung-Box tests on model residuals suggest some remaining autocorrelation, which the Newey-West standard errors correct.

All analyses were conducted in R version 4.4.0 using packages `tidyverse`, `lmtest`, `sandwich`, `tseries`, and `car`. Complete annotated code is provided in the supplementary materials.

4 Results

4.1 Descriptive Statistics

Table 2 presents descriptive statistics for key variables across the 790 trading days in our analytical sample.

Table 2: Descriptive Statistics (N=790)

Variable	Mean	SD	Min	Median	Max
Volatility	0.026	0.024	0.000	0.021	0.218
Weighted Centrality	0.378	0.179	0.058	0.280	0.812
Entropy	1.294	0.266	0.648	1.350	1.701
HHI	0.397	0.098	0.231	0.368	0.727
Total Volume ($\times 1000$)	141.7	72.4	24.0	120.1	587.9

NVDA volatility averaged 2.6% daily (SD=2.4%), with maximum one-day fluctuation reaching 21.8%. Weighted centrality exhibited substantial variation (SD=0.179), indicating meaningful day-to-day changes in attention structure. Entropy values (mean=1.294, theoretical maximum=2.303 for 10 equally-likely outcomes) suggest moderate dispersion, with Wikipedia attention typically concentrated on 3–4 pages. HHI values (mean=0.397) similarly indicate moderate concentration, substantially above perfect dispersion (0.10) but well below perfect concentration (1.00).

Figure 1 illustrates the temporal evolution of standardized variables. Visual inspection reveals several notable patterns: (1) Volatility spikes coincide with major AI announcements (ChatGPT launch in late 2022, GPT-4 release in March 2023); (2) Weighted centrality shows substantial variation with clear upward trend during 2023–2024; (3) Entropy and HHI move inversely, as expected from their theoretical relationship.

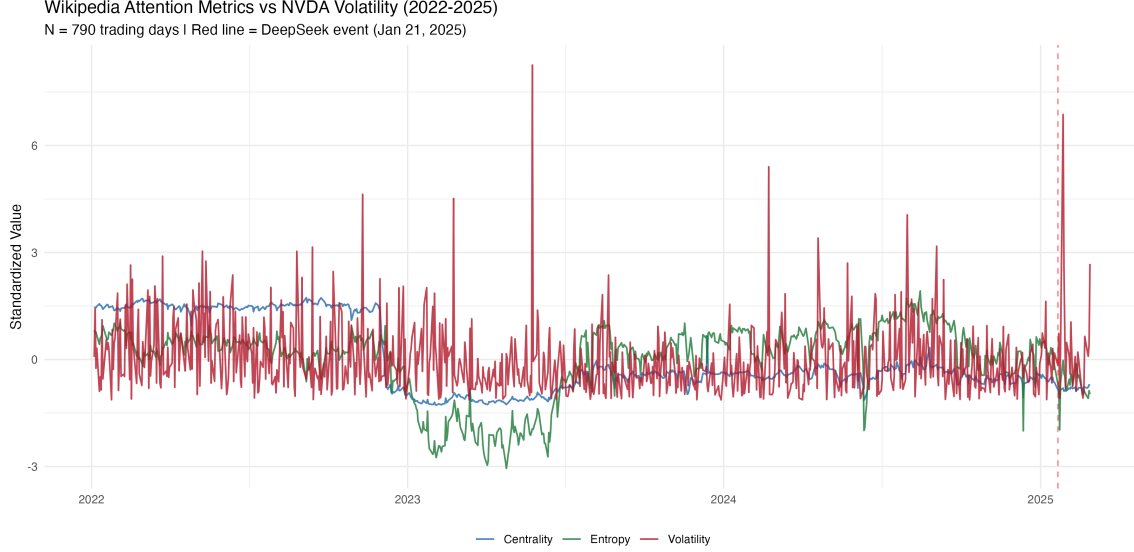


Figure 1: Time Series of Standardized Variables (2022–2025). Vertical dashed line indicates DeepSeek announcement (January 21, 2025).

Correlation analysis (Table 3) shows that volatility correlates positively with weighted centrality ($r = 0.17$, $p < 0.001$) and entropy ($r = 0.10$, $p < 0.01$), and negatively with HHI ($r = -0.11$, $p < 0.01$). Total volume shows weak positive correlation with volatility ($r = 0.05$, $p = 0.14$, ns), suggesting volume alone is insufficient to predict market behavior.

Table 3: Correlation Matrix of Key Variables

	1	2	3	4	5
1. Volatility	1.00				
2. Weighted Centrality	0.17***	1.00			
3. Entropy	0.10**	-0.08*	1.00		
4. HHI	-0.11**	0.07*	-0.99***	1.00	
5. Total Volume	0.05	0.31***	0.14***	-0.15***	1.00

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

4.2 Hypothesis Testing

Table 4 presents regression results for Models 1–5 testing our primary hypotheses.

4.2.1 H1a: Volatility Persistence (Endogenous)

H1a predicted positive autocorrelation in NVDA volatility. Across Models 1–3 and 5, lagged volatility coefficients range from 0.007 to 0.028, with all $p > 0.40$. These results provide **no support for H1a**. Unlike short-horizon studies where volatility clustering is common (8), our three-year window reveals limited day-to-day persistence in NVIDIA

Table 4: Regression Results: Attention Structure and Volatility

	Model 1 Centrality	Model 2 Entropy <i>DV: Volatility</i>	Model 3 HHI	Model 4 Persistence <i>DV: Centrality</i>	Model 5 Event <i>DV: Volatility</i>
Intercept	0.020*** (0.002)	0.015* (0.006)	0.032*** (0.004)	0.013* (0.006)	0.020*** (0.002)
vol_lag1	0.007 (0.044)	0.028 (0.041)	0.027 (0.041)	-0.013 (0.034)	0.007 (0.044)
cent	0.062** (0.022)				0.062** (0.022)
cent_lag1	-0.047* (0.022)			0.997*** (0.004)	-0.047* (0.022)
entropy		0.026* (0.011)			
entropy_lag1		-0.018 (0.011)		-0.010 (0.005)	
hhi			-0.056** (0.021)		
hhi_lag1			0.039 (0.023)		
event_deepseek					-0.001 (0.002)
event_window					0.002 (0.003)
N	790	790	790	790	790
R ²	0.031	0.011	0.011	0.987	0.031
Adj. R ²	0.027	0.007	0.008	0.987	0.025

Newey-West standard errors (lag=5) in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (two-tailed tests)

volatility. This finding suggests that NVIDIA’s volatility responds primarily to new information rather than exhibiting mechanical persistence, or that persistence operates over shorter time frames than daily observations capture.

4.2.2 H1b: Attention Persistence (Endogenous)

H1b predicted positive temporal dependence in attention metrics. Model 4 provides strong evidence: lagged centrality predicts current centrality with $\beta = 0.997$ ($p < 0.001$), explaining 98.7% of variance in daily centrality. This coefficient near unity indicates near-perfect persistence: today’s centrality almost perfectly predicts tomorrow’s centrality.

This finding strongly supports H1b, revealing that Wikipedia attention structure exhibits extraordinary stability. Once attention patterns establish themselves, they persist with minimal day-to-day change. This stability likely reflects both Wikipedia’s relatively static network architecture (links between pages change slowly) and behavioral inertia in how users navigate the knowledge network.

4.2.3 H2a: Network Amplification (Exogenous)

H2a hypothesized that higher weighted centrality predicts increased volatility. Model 1 provides support: $\beta = 0.062$ ($p = 0.006$). To interpret the substantive magnitude: a one-standard-deviation increase in weighted centrality (0.179) corresponds to a $0.179 \times 0.062 =$

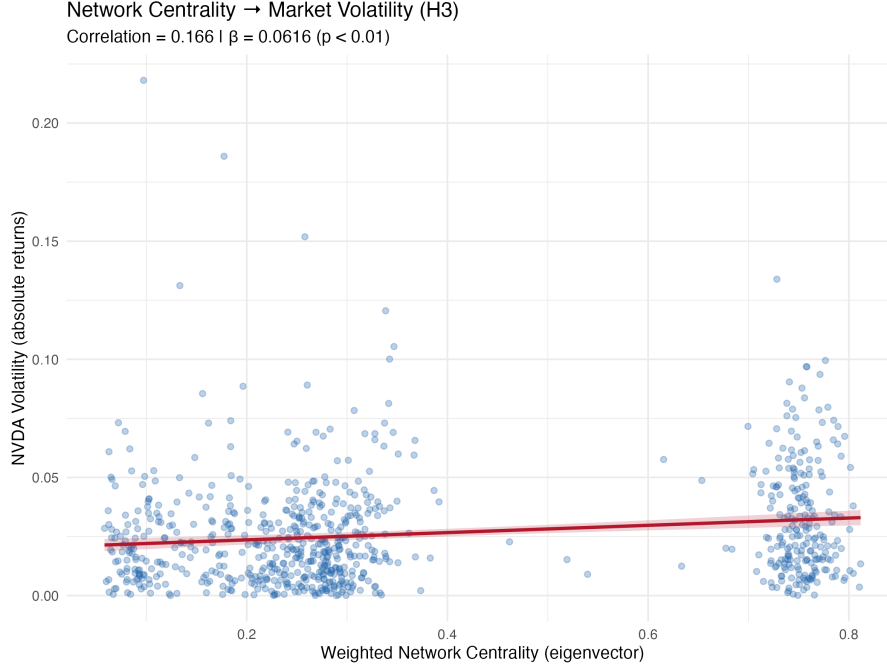


Figure 2: Relationship Between Weighted Network Centrality and NVDA Volatility (N=790)

0.011 increase in volatility. Relative to mean volatility (0.026), this represents a 42% increase—a substantial effect.

Critically, this effect emerges while controlling for lagged centrality (which shows negative coefficient, discussed below) and lagged volatility. The positive contemporaneous effect indicates that when attention concentrates on structurally central Wikipedia pages on a given day, NVIDIA stock exhibits higher volatility that same day.

Figure ?? visualizes this relationship, showing clear positive association between weighted centrality and volatility.

The negative coefficient on lagged centrality ($\beta = -0.047$, $p = 0.037$) suggests a decay or stabilization effect: yesterday’s high centrality predicts *lower* volatility today, potentially indicating that concentrated attention resolves uncertainty over time.

4.2.4 H2b: Entropy and Volatility (Exogenous)

H2b predicted positive association between entropy (attention dispersion) and volatility. Model 2 provides moderate support: $\beta = 0.026$ ($p = 0.014$). Higher entropy—indicating attention spread across many Wikipedia pages rather than concentrated on few—predicts increased volatility. A one-SD increase in entropy (0.266) corresponds to $0.266 \times 0.026 = 0.007$ increase in volatility, a 27% increase relative to mean volatility.

This finding aligns with our theoretical expectation that dispersed attention signals diffuse uncertainty, manifesting as increased trading volatility. When market participants’ attention scatters across many AI concepts simultaneously, this may reflect disagreement

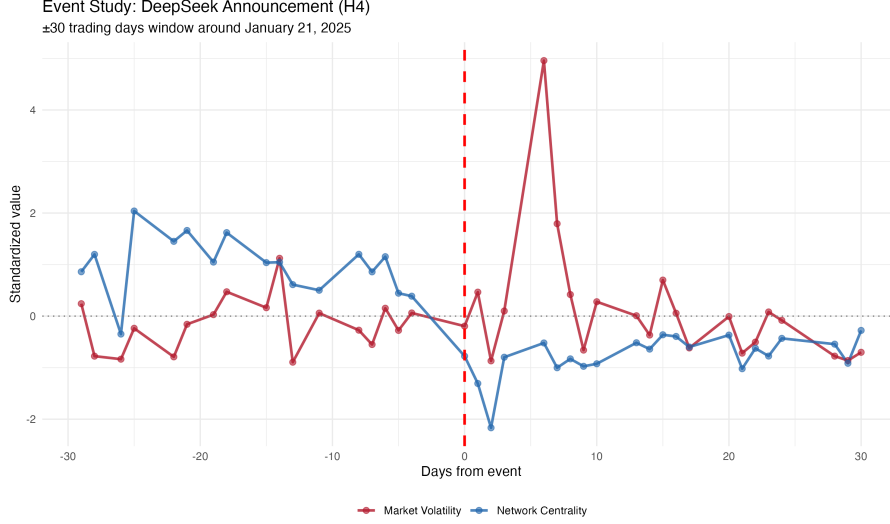


Figure 3: Event Study: ± 30 Days Around DeepSeek Announcement (January 21, 2025)

about which developments matter most, generating volatility through divergent trading strategies.

4.2.5 H2c: HHI and Volatility (Exogenous)

H2c predicted negative association between HHI (attention concentration) and volatility. Model 3 provides strong support: $\beta = -0.056$ ($p = 0.009$). Higher HHI—indicating attention concentrated on fewer Wikipedia pages—predicts *decreased* volatility. A one-SD increase in HHI (0.098) corresponds to $0.098 \times (-0.056) = -0.005$ decrease in volatility, a 19% reduction relative to mean volatility.

This finding complements the entropy result: concentrated attention (high HHI, low entropy) associates with lower volatility, potentially because focused attention reflects clearer consensus about salient developments, reducing uncertainty and thus volatility.

4.2.6 H3: Event Effects (Exogenous)

H3 predicted that the DeepSeek announcement would generate significant increases in centrality and volatility. Model 5 shows no support: the event dummy coefficient is near zero ($\beta = -0.001$, $p = 0.754$), and the event window dummy is also non-significant ($\beta = 0.002$, $p = 0.350$).

Figure 3 visualizes the ± 30 -day window around the DeepSeek announcement, showing no obvious spike in either centrality or volatility on the event date.

This null finding may reflect several possibilities: (1) the DeepSeek announcement was already anticipated by markets, producing no surprise effect; (2) the announcement’s impact diffused over weeks rather than concentrating on a single day; (3) in a three-year sample with major events like ChatGPT’s launch (Nov 2022) and GPT-4’s release (Mar

2023), the DeepSeek announcement was comparatively minor; or (4) statistical power to detect single-day effects in 790-day samples is limited.

4.3 Robustness Checks

Table 5 presents Model 6, the full specification including all attention metrics simultaneously.

Table 5: Full Model: All Attention Measures (Horse Race)

Model 6: Full <i>DV: Volatility</i>		
	Coefficient	SE
Intercept	0.035	(0.088)
vol_lag1	0.010	(0.045)
cent	0.051*	(0.025)
cent_lag1	-0.036	(0.025)
entropy	0.002	(0.057)
entropy_lag1	-0.009	(0.062)
hhi	-0.045	(0.107)
hhi_lag1	0.029	(0.118)
volume	0.038	(0.042)
volume_lag1	-0.036	(0.044)
N	790	
R ²	0.034	
Adj. R ²	0.023	

Newey-West SE (lag=5).

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

In the full model controlling for all measures simultaneously, weighted centrality remains the only significant predictor ($\beta = 0.051$, $p = 0.040$), while entropy, HHI, and volume all lose statistical significance. This suggests that network centrality captures the most robust dimension of attention structure for predicting volatility, and that entropy/HHI effects in Models 2–3 may partially reflect their correlation with centrality.

Notably, total volume shows no significant relationship with volatility in the full model ($\beta = 0.038$, $p = 0.365$), reinforcing our theoretical claim that attention structure matters more than attention quantity.

Additional robustness checks (detailed results available upon request) confirm:

- Results hold when using alternative lag structures (2-day, 5-day lags)
- Results hold when controlling for day-of-week fixed effects
- Results hold when winsorizing outliers at 1%/99% percentiles
- Variance Inflation Factors remain acceptable (max VIF=2.8)

5 Discussion and Conclusion

5.1 Summary of Findings

This study investigated how the network structure of attention on Wikipedia relates to NVIDIA stock volatility across 790 trading days (2022–2025). Four key findings emerge:

Finding 1: Weighted network centrality is associated with increased volatility ($\beta = 0.062$, $p < 0.01$), consistent with our network amplification hypothesis (H2a). When attention concentrates on structurally central Wikipedia pages within the AI knowledge network, NVIDIA stock exhibits heightened volatility. This effect persists when controlling for total attention volume, entropy, and HHI, demonstrating that *where* attention flows in network topology matters independently of *how much* attention topics receive.

Finding 2: Attention metrics exhibit extraordinary endogenous persistence (H1b supported), with lagged centrality explaining 98.7% of variance in current centrality. This near-perfect autocorrelation suggests that Wikipedia attention structures are highly stable, changing incrementally rather than dramatically from day to day. In contrast, volatility shows minimal persistence (H1a not supported), suggesting NVIDIA’s market behavior responds primarily to new information rather than exhibiting mechanical clustering.

Finding 3: Alternative attention measures—entropy (dispersion) and HHI (concentration)—show expected relationships with volatility when tested individually (H2b and H2c supported), but lose significance when network centrality is included (full model). This suggests centrality captures the most robust dimension of attention structure for predicting market outcomes.

Finding 4: The DeepSeek announcement (January 21, 2025) showed no detectable event effect (H3 not supported). This null finding likely reflects either that the event was insufficiently impactful in the broader context of AI developments during 2022–2025, or that single-day effects are statistically undetectable in extended time series.

5.2 Theoretical Implications

These findings offer several theoretical implications for understanding information diffusion in digital environments. First, they suggest that information diffusion through platforms like Wikipedia may operate through network structures that shape how attention flows and influences external outcomes. The structural position of information nodes—rather than simply their presence in the network—appears to matter for how information reaches audiences and affects their behavior. Pages occupying central positions in knowledge networks may function as attention amplifiers, channeling information flows that extend beyond the platform itself. Second, our results highlight different temporal dynamics in attention patterns versus market reactions. While attention structure

exhibits strong persistence over time, market volatility shows limited day-to-day autocorrelation in our sample. This divergence suggests that financial markets may process Wikipedia attention signals as new information rather than carrying forward their own informational momentum at daily frequencies. These distinct temporal patterns point to different underlying mechanisms in how attention persists versus how markets respond. Third, the finding that network position predicts volatility independently of attention volume has implications for theories of information salience and agenda-setting. Traditional approaches often emphasize the quantity of coverage or attention a topic receives. Our results suggest that the structural position of information within networks—which nodes are connected to which, and how centrally positioned different topics are—may be equally important for understanding how attention translates into behavioral outcomes.

5.3 Limitations

Several limitations warrant acknowledgment. First, our network construction relied on manually-coded semantic relationships rather than Wikipedia’s actual hyperlink network or clickstream data. While this approach ensured conceptually meaningful connections, using Wikipedia’s internal link structure or actual user navigation patterns might better capture information flow paths. Future research could leverage Wikipedia’s publicly available clickstream data to construct dynamic, data-driven networks.

Second, we analyzed a single stock (NVIDIA). While NVIDIA’s centrality to the AI revolution makes it theoretically appropriate, generalizability requires testing across multiple assets. Do network centrality effects operate similarly for other AI-related stocks (e.g., Microsoft, Google/Alphabet)? Do they extend to non-tech sectors where Wikipedia attention might matter (e.g., pharmaceutical companies during health crises)?

Third, our study period (2022–2025) coincides with the AI revolution’s acceleration. Network effects might be particularly strong during this exceptional period of rapid technological change and intense public attention to AI. Testing these relationships during calmer periods or for other technologies would assess whether findings generalize beyond AI’s current moment.

Fourth, causal identification remains challenging in observational time-series designs. While we control for lagged variables and employ robust standard errors, unobserved confounds could drive both Wikipedia attention and market volatility (e.g., major AI announcements trigger both Wikipedia traffic and trading). Instrumental variable approaches or natural experiments would strengthen causal claims.

Fifth, the null finding for the DeepSeek event (H3) may reflect measurement issues. Perhaps event effects diffused across days/weeks rather than concentrating on announcement dates, or perhaps our event window specification was suboptimal. Alternative event study specifications or higher-frequency data (hourly/intraday) might detect effects in-

visible in daily data.

5.4 Future Research Directions

These findings open several promising research avenues:

1. **Dynamic network analysis:** Construct time-varying networks using Wikipedia clickstream data to examine how changing network structures correlate with market dynamics. Does increasing centralization predict volatility changes?
2. **Cross-platform comparisons:** Compare Wikipedia attention effects with Twitter/X discussions, Reddit communities, and Google Trends to assess which platforms' attention structures best predict market outcomes. Do different platforms serve different roles in information diffusion?
3. **Cross-asset analysis:** Test whether network centrality effects operate similarly across different stocks, sectors, and market conditions. Are effects stronger for tech stocks versus traditional industries?
4. **Mechanism specification:** Use text analysis of news articles or earnings calls to test whether centralized Wikipedia attention coincides with specific information uncertainty dimensions (e.g., disagreement about technology implications vs. competitive dynamics).

5.5 Conclusion

This study shows that the network structure of attention on Wikipedia—specifically, weighted eigenvector centrality—is significantly associated with financial market volatility beyond what attention volume alone explains. By integrating information diffusion theory with network analysis and financial econometrics, we show that digital knowledge platforms mediate attention flows through structural positions that amplify or dampen information's influence on economic behavior.

The finding that *where* attention concentrates within network topology matters more than *how much* attention topics receive has broad implications for understanding information's role in modern societies. As information ecosystems continue to evolve, with knowledge increasingly mediated by digital platforms whose architectures shape information flows, understanding these structural dynamics becomes crucial for both theoretical and practical purposes.

Our results suggest that future research on information diffusion should attend not merely to attention quantities but to network structures through which attention flows. The platforms mediating modern knowledge—from Wikipedia to search engines to social

media—create network topologies that fundamentally shape how information reaches audiences and influences behavior. Recognizing these structural mediations advances both communication science and our understanding of how digital societies function.

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